In [2]:
```
# import libs
import pandas as pd

# load data
raw_dataset = pd.read_csv("../data/1_raw_dataset.csv", low_memory=False)
```

Dataset engineering

To view the final variables used in the dataset, click here

Preface

The treatment realised here are being done after the construction of our rudimentary dataset.

This dataset is the merger of the following sources :

- TTC Delays and Routes 2023 - Kaggle 2023 (2025-10-15)
- Hourly climate data (station 6158359) - Government of Canada (2025-10-15)

To each entry in the TTC delay source, we associated the hourly weather data. The association is made from the local hour (UTC−4) of the delay rounded up to the nearest hour (floored <= 30min, ceiled > 30min)

Objective

Our final objective it to predict the delay (at the scale of the minute) on a bus line at a precise time in relation to the local weather conditions.

Our goal here is to remove all the variables that are not of use to reach this objective.

Initial variables description

The table just below details all the different variables present in our dataset.

Name	Type	Origin	Description
Date	Date	Kaggle 2023	The date when the delay incident occurred (format: 'd-MMM-yy', UTC−4)
Route	Nominal categorical	Kaggle 2023	The bus line number

Name	Type	Origin	Description
Time	Time (HH:MM)	Kaggle 2023	The time when the delay incident was reported in the 24h format (UTC-4)
Day	Nominal categorical	Kaggle 2023	The day of the week when the delay incident occurred
Location	Nominal categorical	Kaggle 2023	The location or station where the delay incident occurred
Incident	Nominal categorical	Kaggle 2023	The type of incident causing the delay
Min Delay	**Discrete numerical**	Kaggle 2023	**The duration of the delay in minutes**
Min Gap	Discrete numerical	Kaggle 2023	The time gap in minutes between buses
Direction	Nominal categorical	Kaggle 2023	The direction of travel
Vehicle	Nominal categorical	Kaggle 2023	The identifier of the bus involved in the delay incident
x	Continuous numerical	Government of Canada	Longitude of the weather station
y	Continuous numerical	Government of Canada	Latitude of the weather station
STATION_NAME	Text	Government of Canada	Name of the weather station
CLIMATE_IDENTIFIER	Text	Government of Canada	Identifier of the weather station
ID	Text	Government of Canada	Identifier of the weather measure
LOCAL_DATE	Date	Government of Canada	Date of the weather measure (format: 'YYYY-MM-DD HH:MM:SS': UTC-4)
PROVINCE_CODE	Nominal categorical	Government of Canada	Code of the province in which the measure was made
LOCAL_YEAR	Discrete numerical	Government of Canada	Local year during which the measure was made (UTC-4)
LOCAL_MONTH	Nominal categorical	Government of Canada	Local month during which the measure was made (note: is encoded from 1-12 as floats, UTC-4)
LOCAL_DAY	Nominal categorical	Government of Canada	Local day of the month during which the measure was made (note: is encoded from 1-31 as floats, UTC-4)

Name	Type	Origin	Description
LOCAL_HOUR	Nominal categorical	Government of Canada	Local hour of the day during which the measure was made (note: encoded from 0-23 as floats & the minutes are always at 00, UTC-4)
UTC_DATE	Date	Government of Canada	Date of the weather measure (format: 'YYYY-MM-DDTHH:MM:SS': UTC+0)
UTC_YEAR	Discrete numerical	Government of Canada	UTC year during which the measure was made (UTC+0)
UTC_MONTH	Nominal categorical	Government of Canada	UTC month during which the measure was made (note: is encoded from 1-12 as floats, UTC+0)
UTC_DAY	Nominal categorical	Government of Canada	UTC day of the month during which the measure was made (note: is encoded from 1-31 as floats, UTC+0)
UTC_HOUR	Nominal categorical	Government of Canada	UTC hour of the day during which the measure was made (note: encoded from 0-23 as floats & the minutes are always at 00, UTC+0)
TEMP	Continuous numerical	Government of Canada	Temperature in degrees Celsius
TEMP_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the temperature (FLAG TABLE)
DEW_POINT_TEMP	Continuous numerical	Government of Canada	Dew point temperature in degrees Celsius
DEW_POINT_TEMP_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the dew point temperature (FLAG TABLE)
HUMIDEX	Continuous numerical	Government of Canada	Index indicating how hot or humid the weather feels to the average person
HUMIDEX_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the humidex (FLAG TABLE)
PRECIP_AMOUNT	Continuous numerical	Government of Canada	Measurment of precipitation expressed in terms of vertical depths of water in millimeter (mm)
PRECIP_AMOUNT_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the preciîtation (FLAG TABLE)
RELATIVE_HUMIDITY	Continuous numerical	Government of Canada	Relative humidity in percent (%)

Name	Type	Origin	Description
RELATIVE_HUMIDITY_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the relative humidity (FLAG TABLE)
STATION_PRESSURE	Continuous numerical	Government of Canada	The atmospheric pressure in kilopascals (kPa) at the station elevation
STATION_PRESSURE_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the atmospheric pressure (FLAG TABLE)
VISIBILITY	Continuous numerical	Government of Canada	The distance in kilometers (km) at which objects of suitable size can be seen and identified
VISIBILITY_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the visibility (FLAG TABLE)
WEATHER_ENG_DESC	Nominal categorical	Government of Canada	Observation of atmospheric phenomenon including the occurrence of weather and obstructions to vision in english (note: no data means "Clear")
WEATHER_FRE_DESC	Nominal categorical	Government of Canada	Observation of atmospheric phenomenon including the occurrence of weather and obstructions to vision in french
WINDCHILL	Continuous numerical	Government of Canada	Index indicating how cold the weather feels to the average person
WINDCHILL_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the windchill (FLAG TABLE)
WIND_DIRECTION	Continuous numerical	Government of Canada	The geographic direction from which the wind blows expressed in tens of degrees
WIND_DIRECTION_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the wind direction (FLAG TABLE)
WIND_SPEED	Continuous numerical	Government of Canada	Measure of the wind speed in kilometers per hour (km/h). Usually observed at 10 meteres above the ground
WIND_SPEED_FLAG	Nominal categorical	Government of Canada	Flag for the measure of the wind speed (FLAG TABLE)

Removing variables

At first glance, we can see that some explanatory variables cannot be used for the analysis (not usable in our final application, metadata, duplicates, etc.).

The table below describes what variables are removed from the dataset.

Name	Reason	Action	Explanation
Date	Duplicate	Remove	The date is already splitted in all it's basic components in our dataset
Location	Usecase	Temporary removal	During the prediction, we will not have access to the incident location. Howerver we could use this data to have the delay per station later on
Min Gap	Duplicate	Remove	This variable is another representation of our target variable
Direction	Usecase	Remove	During the prediction, we will not have access to the direction of the buses
Vehicle	Usecase/Unusable	Remove	During the prediction, we will not have access to the current buses circulating. Furthermore, the buses may have changed today, rendering this variable unusable
x	Metadata	Remove	The longitude of the weather station is irrelevant to our predictions
y	Metadata	Remove	The latitude of the weather station is irrelevant to our predictions
STATION_NAME	Metadata	Remove	The name of the weather station is irrelevant to our predictions
CLIMATE_IDENTIFIER	Metadata	Remove	The ID of the weather station is irrelevant to our predictions
ID	Metadata	Remove	The ID of the weather prediction is irrelevant to our predictions
LOCAL_DATE	Duplicate	Remove	The date is already splitted in all it's basic components in our dataset
PROVINCE_CODE	Metadata	Remove	The province code is irrelevant to our predictions
LOCAL_YEAR	Unusable	Remove	The year of the accident will not repeat itself. It is irrelevant for our predictions
LOCAL_HOUR	Duplicate	Remove	The "Time" variable describes more precisely the hour and minute of the incident (note: this variable actually represents the hour of the measurement of the weather, not the time of the incident)
UTC_*	Duplicate	Remove	All the UTC_* variables are duplicates of the local time variables. However

Name	Reason	Action	Explanation
			our incident data is based on the local hour (UTC-4)
WEATHER_FRE_DESC	Duplicate	Remove	Is a duplicate of the same english variable
*_FLAG	Metadata	Remove	The conditions surrounding the measurment of the weather are not relevant to our predictions. However, this variable is useful to evaluate the quality of the measurments

We proceed to the removal of these variables.

In [3]:
```
var_to_remove = [
    "Date",
    "Location",
    "Min Gap",
    "Direction",
    "Vehicle",
    "x",
    "y",
    "STATION_NAME",
    "CLIMATE_IDENTIFIER",
    "ID",
    "LOCAL_DATE",
    "PROVINCE_CODE",
    "LOCAL_YEAR",
    "LOCAL_HOUR",
    "UTC_DATE", "UTC_YEAR", "UTC_MONTH", "UTC_DAY",
    "WEATHER_FRE_DESC",
    "TEMP_FLAG", "DEW_POINT_TEMP_FLAG", "HUMIDEX_FLAG", "PRECIP_AMOUNT_FLAG", "RELATI
    "VISIBILITY_FLAG", "WINDCHILL_FLAG", "WIND_DIRECTION_FLAG", "WIND_SPEED_FLAG"
]

dataset = raw_dataset.drop(columns=var_to_remove)

dataset.head()
```

Out[3]:

	Route	Time	Day	Incident	Min Delay	LOCAL_MONTH	LOCAL_DAY	TEMP	DEW_POI
0	91	02:30	Sunday	Diversion	81	1.0	1.0	3.7	
1	69	02:34	Sunday	Security	22	1.0	1.0	3.5	
2	35	03:06	Sunday	Cleaning - Unsanitary	30	1.0	1.0	3.5	
3	900	03:14	Sunday	Security	17	1.0	1.0	3.5	
4	85	03:43	Sunday	Security	1	1.0	1.0	4.4	

Renaming variables

Since we mixed two datasets, the naming of the variables is not uniform. We will rename the variables to make them more readable.

Old name	New name
Route	ROUTE
Time	LOCAL_TIME
Day	WEEK_DAY
Incident	INCIDENT
Min Delay	DELAY

In [4]:
```
dataset = dataset.rename(columns={
    "Route": "ROUTE",
    "Time": "LOCAL_TIME",
    "Day": "WEEK_DAY",
    "Incident": "INCIDENT",
    "Min Delay": "DELAY"
})

dataset.head()
```

Out[4]:

	ROUTE	LOCAL_TIME	WEEK_DAY	INCIDENT	DELAY	LOCAL_MONTH	LOCAL_DAY	TEM
0	91	02:30	Sunday	Diversion	81	1.0	1.0	3.
1	69	02:34	Sunday	Security	22	1.0	1.0	3.
2	35	03:06	Sunday	Cleaning - Unsanitary	30	1.0	1.0	3.
3	900	03:14	Sunday	Security	17	1.0	1.0	3.
4	85	03:43	Sunday	Security	1	1.0	1.0	4.

Final variables

Here are the variables currently present in our dataset.

WARNING : this dataset is not final. During the EDA we will continue to modify this dataset before feeding it to a predictive model.

Name	Type	Origin	Description
ROUTE	Nominal categorical	Kaggle 2023	The bus line number
LOCAL_TIME	Time (HH:MM)	Kaggle 2023	The time when the delay incident was reported in the 24h format (UTC-4)
WEEK_DAY	Nominal categorical	Kaggle 2023	The day of the week when the delay incident occurred
INCIDENT	Nominal categorical	Kaggle 2023	The type of incident causing the delay
DELAY	**Discrete numerical**	Kaggle 2023	**The duration of the delay in minutes**
LOCAL_MONTH	Nominal categorical	Government of Canada	Local month during which the measure was made (note: is encoded from 1-12 as floats, UTC-4)
LOCAL_DAY	Nominal categorical	Government of Canada	Local day of the month during which the measure was made (note: is encoded from 1-31 as floats, UTC-4)
TEMP	Continuous numerical	Government of Canada	Temperature in degrees Celsius
DEW_POINT_TEMP	Continuous numerical	Government of Canada	Dew point temperature in degrees Celsius (temperature at which water vapor contained in the air condenses at contact with a cold surface)
HUMIDEX	Continuous numerical	Government of Canada	Index indicating how hot or humid the weather feels to the average person
PRECIP_AMOUNT	Continuous numerical	Government of Canada	Measurment of precipitation expressed in terms of vertical depths of water in millimeter (mm)
RELATIVE_HUMIDITY	Continuous numerical	Government of Canada	Relative humidity in percent (%)
STATION_PRESSURE	Continuous numerical	Government of Canada	The atmospheric pressure in kilopascals (kPa) at the station elevation
VISIBILITY	Continuous numerical	Government of Canada	The distance in kilometers (km) at which objects of suitable size can be seen and identified
WEATHER_ENG_DESC	Nominal categorical	Government of Canada	Observation of atmospheric phenomenon including the occurrence of weather and obstructions to vision in english
WINDCHILL	Continuous numerical	Government of Canada	Index indicating how cold the weather feels to the average person

Name	Type	Origin	Description
WIND_DIRECTION	Continuous numerical	Government of Canada	The geographic direction from which the wind blows expressed in tens of degrees
WIND_SPEED	Continuous numerical	Government of Canada	Measure of the wind speed in kilometers per hour (km/h). Usually observed at 10 metres above the ground

We now save this new dataset under the name `"2_dataset.csv"`.

In [5]:
```
dataset.to_csv("../data/2_dataset.csv", index=False)
```